

The Codex.

IAQ's business, easy to read.

Three sections for three readers. Section 1 is for anyone. Section 2 is for engineers and buyers who know the industry. Section 3 is how the website shows it.

How to read every picture: the six kinds of thing.

One colour and one icon per kind, the same everywhere on this page. Summary tells them apart; Detailed says what each one is, what IAQ calls it, and what it is not.

**Business unit**
Three

The company inside IAQ that signs the contract and answers for the result. A buyer chooses the unit first, because the unit decides who is accountable.

**EPC**
Engineering, Procurement and Construction. Builds the facility.

**PCU & TTI**
Process Critical Utilities & Total Tool Installation. Re-equips a live semiconductor fab.

**EFM**
Energy Facility Management. Runs and maintains it, and cuts the energy bill.

“We bought the whole fab from the EPC unit. Two years later EFM took over the chiller plant.”

Think
Not
On the site

The company you sign with.

Not a service. A unit sells several services; “EPC” is a unit and also the name of a contract model, which is why it gets confused.

The first group of the Services menu, and the three unit pages.

**Delivery model**
Four

How the contract runs: who holds the construction contracts, who answers for the system performing, and how IAQ is paid. Every model belongs to one unit.

**EPCC**
IAQ delivers everything, commissioning included, and answers for the result. EPC unit.

**EPCM**
The owner holds the contracts, IAQ manages them. EPC unit.

**Cooling as a Service**
IAQ funds and runs the chiller plant for a fixed tariff. EFM.

**Energy Performance Contracting**
IAQ funds the upgrade and is paid from the measured savings. EFM.

“Same six services, but on EPCM the owner keeps the contracts and IAQ manages the programme.”

Think
Not
On the site

The shape of the deal.

Not a service and not a unit. It changes who is accountable, not what gets built.

Grey chips under each unit in the Services menu; the four-ways slide; each unit page.

**Service**
Six

What is written into the contract, in the order a project runs. Any one can be bought alone, or several, or all six as one cycle.

**1 Design**
Engineering design and consultation.

**2 Procure**
Procurement.

**3 Construct**
Construction.

**4 Commission**
Testing and commissioning.

**5 Maintain**
Maintenance.

**6 Hook up**
Tools hookup: production tools connected and released.

“Design only, please. We already hold the rest of the scope.”

Think
Not
On the site

The menu.

Not a system. Process cooling water is a system that two services touch; it is never a service itself.

The six-service strip in the Services menu, the delivery cycle, and one page per service.



Work

Three

The discipline that does the work on site, which is how the market classifies a contractor. Most contractors carry one; IAQ carries all three in house.



CSA

Civil, Structural and Architectural: the building itself.



MEP

Mechanical, Electrical and Plumbing: what makes it run.



Process utilities

The utilities a production tool consumes: gases, chemicals, pure water, exhaust.

“A civil contractor does CSA only. IAQ does CSA, MEP and process, so the whole scope stays with one firm.”

Think The trades on site.

Not Not a service. A service is what you buy; the work is who does it and how they are classified.

On the site The work layer on every unit page, and the three-disciplines slide.



System

Thirteen listed

The physical thing being built, always inside one kind of work. A system is what the drawings show and what commissioning proves.



Process cooling water

MEP.



Ultrapure water

Process utilities.



Fan filter units

MEP.



Fire protection

MEP.



Specialty gases

Process utilities.



Cleanroom envelope

CSA.

“The PCW loop is built in MEP work, then the last metres to each tool are hookup.”

Think The parts of the fab.

Not Not a service, even when it is priced on its own. “Buy the PCW package” is a scope of construction, not a seventh service.

On the site Green chips under each kind of work on the unit pages; the fab model on the home page.



Market

Seven

Who the facility is for. Each market measures a facility by something different, so it decides the class or spec IAQ builds to.



Semiconductor

Particle count, ISO 3 to 7.



Data centre

Uptime and thermal stability.



EV battery

Dew point, dry room.



Photovoltaics

Process stability at scale.



District cooling

Kilowatt hours.



Bio LifeScience

GMP grade.



Food & Beverage

Hygiene regime.

“An EV battery plant is judged by dew point, so the dry room is the spec that matters.”

Think The customer.

Not Not a unit and not a service. IAQ serves every market with the same three units.

On the site The Markets menu, the home page market rail, one page per market.

Side by side, **what each one really is.**

Compared	 Business unit	 Delivery model	 Service	 Work	 System	 Market
What it is	The company inside IAQ that signs the contract and answers for the result. A buyer chooses the unit first, because the unit decides who is accountable.	How the contract runs: who holds the construction contracts, who answers for the system performing, and how IAQ is paid. Every model belongs to one unit.	What is written into the contract, in the order a project runs. Any one can be bought alone, or several, or all six as one cycle.	The discipline that does the work on site, which is how the market classifies a contractor. Most contractors carry one; IAQ carries all three in house.	The physical thing being built, always inside one kind of work. A system is what the drawings show and what commissioning proves.	Who the facility is for. Each market measures a facility by something different, so it decides the class or spec IAQ builds to.
How many IAQ has	Three	Four	Six	Three	Thirteen listed	Seven
The names	EPC · PCU & TTI · EFM	EPCC · EPCM · Cooling as a Service · Energy Performance Contracting	1 Design · 2 Procure · 3 Construct · 4 Commission · 5 Maintain · 6 Hook up	CSA · MEP · Process utilities	Process cooling water · Ultrapure water · Fan filter units · Fire protection · Specialty gases · Cleanroom envelope	Semiconductor · Data centre · EV battery · Photovoltaics · District cooling · Bio LifeScience · Food & Beverage
Who decides it	The buyer, first of all.	The buyer, with IAQ, before the contract is written.	The buyer, per contract.	The market, when it classifies a contractor.	The design, then the commissioning test.	Who is paying for the facility.
Think of it as	The company you sign with.	The shape of the deal.	The menu.	The trades on site.	The parts of the fab.	The customer.
Often confused with	Not a service. A unit sells several services; “EPC” is a unit and also the name of a contract model, which is why it gets confused.	Not a service and not a unit. It changes who is accountable, not what gets built.	Not a system. Process cooling water is a system that two services touch; it is never a service itself.	Not a service. A service is what you buy; the work is who does it and how they are classified.	Not a service, even when it is priced on its own. “Buy the PCW package” is a scope of construction, not a seventh service.	Not a unit and not a service. IAQ serves every market with the same three units.
Where you see it on the site	The first group of the Services menu, and the three unit pages.	Grey chips under each unit in the Services menu; the four-ways slide; each unit page.	The six-service strip in the Services menu, the delivery cycle, and one page per service.	The work layer on every unit page, and the three-disciplines slide.	Green chips under each kind of work on the unit pages; the fab model on the home page.	The Markets menu, the home page market rail, one page per market.
Colour in every picture	Blue, filled	Ink outline	Red	Amber	Green	Violet

The icon family, one isometric grammar.

Two to four solids on one grid, three flat tones, one accent in the kind’s colour: red for a service, blue for a unit, yellow for work, green for a system (Bazil, 25 Sep). Drawn by components/Isolcon.jsx; used on the Services page and the map.

Services



Design



Procurement



Construction



Commissioning



Maintenance



Tools hookup

Business units



EPC



PCU & TTI



EFM

Work



CSA



MEP



Process utilities

Records



Plant



Award



Medal



Safety

Systems



Piling



Frame



Envelope



Finishes



Air



Pipework



Chiller



Fire



Electrical



Water



Gases



Vacuum and exhaust

Ten questions to ask IAQ, with our answer on each.

Nothing waits on them. Under each question: what was done meanwhile, and the answer
Brand Method would give if the call were ours.

1

Question for Nabilah

Unit 2 is now “Process Critical Utilities & Total Tool Installation Solutions”, IAQ’s own written name, in place of “Total Tools Hookup Solution”. Confirm?

Meanwhile Applied on the site and the slides on 18 Sep, from the cross-check. Hookup now names service 6 only.

Our answer Confirm it. It is IAQ’s own written name from the unit questionnaire, and with it “hookup” means one thing only, service 6. Two names for one unit was the single most confusing thing on the old site.

4

Question for The unit heads

Which of the six services does each unit carry? Please confirm the table on the six services slide, most of all the ones marked “when asked”.

Meanwhile EPC (Engineering, Procurement and Construction) carries all six. The hookup unit carries five, and maintenance when asked. EFM (Energy Facility Management) carries maintenance and commissioning, and the first three when an upgrade needs them. Read from the review and the questionnaires, not yet confirmed by the unit heads.

Our answer Confirm the table as drawn. It follows the 17 Sep review and the questionnaires word for word, and the only cells with any doubt are the “when asked” ones, which cost nothing if a unit head removes one.

7

Question for Nabilah, management

Which figures will IAQ release for a performance slide like the benchmark’s: headcount by function, the safety rate, order intake?

Meanwhile Only the six counters the site already publishes are used. Revenue stays off, as on the site.

Our answer The six counters the site already publishes, plus safe manhours, which IAQ publishes itself. Revenue and order intake stay off until listing counsel clears them; a number that later has to come down does more harm than a slide without it.

10

Question for Nabilah

When does IAQ supply the capability statement PDF?

Meanwhile The Downloads tab in the member portal holds the button, inert until the file arrives. The three ISO certificates download now.

Our answer Send the current PDF even if it is dated. The portal button goes live the day it lands and the file can be swapped at any time; the button staying inert is what a member notices.

2

Question for Nabilah

IAQ’s papers still say founded 1994 and eight countries; the site says 1995 and seven, on your instruction of 10 Sep. Which is final?

Meanwhile The site keeps 1995 and seven.

Our answer Go by the incorporation certificate, and correct whichever side is wrong, including IAQ’s own website. Until it is checked the site keeps 1995 and seven, because the 10 Sep instruction is the latest word. Count countries by offices, not by projects built, and say “projects in more” where it matters.

5

Question for Nabilah, the BIM (Building Information Modelling) manager

Where do cleanroom systems sit in the work layer?

Meanwhile The envelope (walls, ceiling grid, raised floor) sits under CSA, and the fan filter units under MEP.

Our answer Envelope under CSA (Civil, Structural and Architectural), fan filter units under MEP (Mechanical, Electrical and Plumbing). That is how the Revit model is coded and how tender packages split, so the site should not invent a third home for them.

8

Question for Bazil

Do slides 2 and 5 go onto the public Services page in place of the unit cards and the cycle?

Meanwhile They live in the Codex and export for the profile. The public page keeps its current sections until a responsive version is built.

Our answer Yes, as a responsive rebuild of what they show, not as pictures: the three units with what each carries, the one-picture map and the reading order. That is the Services page as built on 24 Sep; the slides themselves are 16:9 images and belong in the profile.

3

Question for Nabilah

Do the six service pages stay as depth reached from the cycle, or fold into the Services page?

Meanwhile Kept, out of the menu, reached from the cycle diagram only.

Our answer Keep them. Each one is a landing page for a search such as “cleanroom commissioning Malaysia”, and a buyer who wants one service wants one page. Out of the menu is right: the cycle and the hub are the way in.

6

Question for Shazwan, Nabilah

What page size does the company profile use, so the slides are placed at the right ratio?

Meanwhile The set exports at 16:9, 1920 by 1080, as PNG and as one PDF. It reflows to A4 landscape with a margin; a portrait profile needs a second layout.

Our answer 16:9 landscape, 1920 by 1080. The set exports natively at that size, reads on a screen, prints on A4 landscape with a margin, and a portrait profile would mean a second layout of every slide for no gain.

9

Question for Sunny

When does the approved project list arrive?

Meanwhile The proof slide is computed from the 18 published references and grows on its own as the registry does.

Our answer Ask Sunny for it in the description format on the project format page, one project at a time if that is faster. Until then the proof slide computes from the 18 published references, so nothing stalls.

1

For everyone. What IAQ does, in plain words.

IAQ is one company with three businesses. Together they build a high-tech factory, fit it out and keep it running. No industry words needed.

1

It builds the factory.

From the first drawing to a finished, clean facility, under one contract.

EPC (Engineering, Procurement and Construction)

2

It connects the machines.

When production machines arrive in a chip factory, it brings them their gases, chemicals, water and power.

PCU & TTI (Process Critical Utilities & Total Tool Installation)

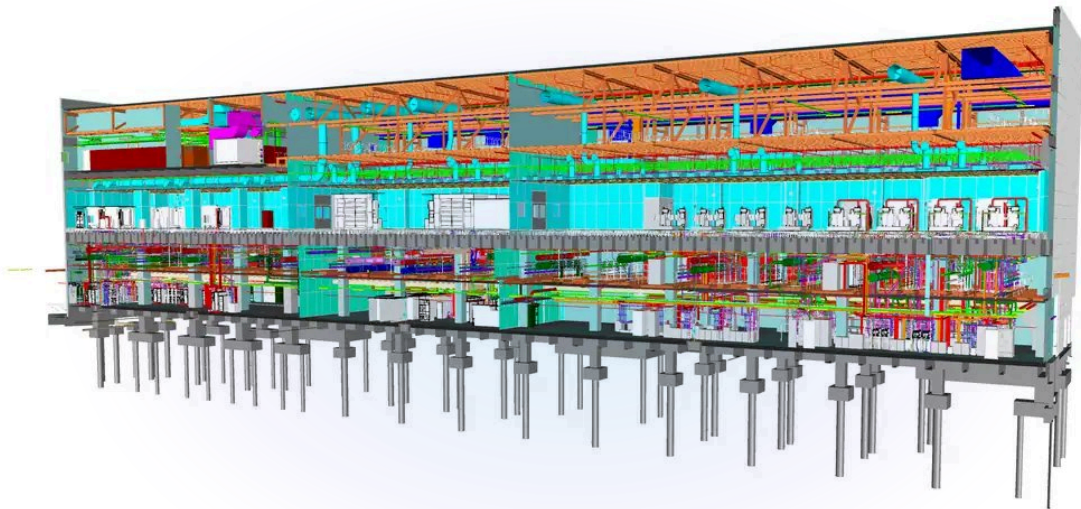
3

It runs it for less.

It maintains the plant and cuts the energy bill, funded by IAQ upfront.

EFM (Energy Facility Management)

Watch a fab being built, and who does each part



1. The ground and the frame

Piles go into the ground. The structure, the floors and the roof go up. **EPC (Engineering, Procurement and Construction)** builds the facility. Work: CSA (Civil, Structural and Architectural). Services: Engineering design, Procurement, Construction.
2. Air, water and power

Ducts, pipes and cables run through every level, so the building can breathe, cool and be powered. **EPC (Engineering, Procurement and Construction)** builds the facility. Work: MEP (Mechanical, Electrical and Plumbing). Services: Construction.
3. The clean room

Sealed walls, a ceiling of air filters and a raised floor keep the air far cleaner than the air outside. **EPC (Engineering, Procurement and Construction)** builds the facility. Work: CSA (Civil, Structural and Architectural), MEP (Mechanical, Electrical and Plumbing). Services: Construction, Testing and commissioning.
4. The machines arrive

The production machines are moved in. Each one is connected to its gases, chemicals, pure water and power, then tested. **PCU & TTI (Process Critical Utilities & Total Tool Installation)** connects the machines. Work: Process utilities, MEP (Mechanical, Electrical and Plumbing). Services: Tools hookup, Testing and commissioning.
5. Running for years

The fab runs day and night. It is maintained, and its energy bill is brought down with no money paid upfront. **EFM (Energy Facility Management)** runs it and cuts the energy bill. Work: MEP (Mechanical, Electrical and Plumbing). Services: Maintenance.

The life of a facility, and who to call at each point

1

Design and build

12 to 24 months

From the first drawing to handover, one party accountable.

Call EPC (Engineering, Procurement and Construction) when

You are building or extending a facility and want one company to answer for all of it.

2

Tools in, fab live

Per tool set

Utilities to each tool, inside a fab that has not stopped.

Call PCU & TTI (Process Critical Utilities & Total Tool Installation) when

Your fab is running and new tools are coming in, or the gas, chemical, water or exhaust systems need to grow.

3

Run and maintain

5 to 20 years

The plant run under a tariff, or the upgrade paid from savings.

Call EFM (Energy Facility Management) when

Your energy or cooling bill is high and you want it cut, with IAQ funding the work and paid from the saving.

Questions clients ask, answered



Choosing a model

How IAQ is engaged, and who answers for what.

Which model should I choose?

EPCC when you want one party to deliver everything and answer for the result. EPCM when you hold the construction contracts yourself and want IAQ to manage them. Large multinationals engage EPCM on the biggest programmes.

[See EPC →](#)

What is the difference between EPCC and EPCM?

EPCC is EPC plus Commissioning: IAQ delivers the whole project, commissioning included. EPCM is EPC Management: IAQ manages the project to the same standard, and the owner holds the construction contracts.

Can I buy one service on its own?

Yes. Procurement only, design only, or design, procure and construct. The scope changes; the EPC model still governs how it runs.

Who is accountable when something does not work?

Under EPCC, IAQ: one contract, one party, from the first drawing to a facility proven to operate as intended. Under EPCM the owner holds the contracts and IAQ manages them; the standard the work is held to is the same.



Time and cost

How long each engagement runs, and what is paid when.

How long does a facility take?



Typically 12 to 24 months under EPC, depending on the project. Tool hookup is priced and scheduled per tool set.

How is tools hookup priced?



Per tool set. Each set is scoped from the tool maker’s facility requirements, so the price follows the tools you bring in, not the size of the fab.

See PCU & TTI →

What does an energy contract cost up front?



Nothing. Under Cooling as a Service IAQ funds, builds and runs the chiller plant for a fixed tariff over 10 to 20 years. Under Energy Performance Contracting IAQ funds the upgrade and is paid from the measured savings over 5 to 10 years.

See EFM →

How are the savings measured?



Against an audited baseline agreed before the work starts. IAQ is paid from the measured saving, so the saving has to be real and has to last.



A live fab

Bringing tools into a fab that has not stopped.

My fab is live. Can tools be brought in while it keeps running?



That is the PCU & TTI unit. It works in four phases, facilitization, rig-in, hook-up and commissioning, and it carries all six services except maintenance, which it adds when asked.

See PCU & TTI →

What are the four phases of a tool installation?



Facilitization, before the tool arrives: everything upstream of its connection point is built to the tool maker’s requirements. Rig-in: the tool is uncrated in a de-trash airlock, moved along an agreed route, set, levelled and anchored. Hook-up: each utility is tapped from the facility and connected into the tool, to its own standard. Commissioning: the tool is proven, then released to production.

What do Process Critical Utilities cover?



The systems a production tool runs on: clean dry air, specialty gases and chemical delivery, ultrapure water, process vacuum and exhaust, and waste treatment.

Does PCU & TTI work outside semiconductor?



No. Semiconductor only. Other hi-tech facilities are built and equipped by EPC under the same six services.



After handover

Running and maintaining what was built.

What happens after handover?



Maintenance is the fifth service and EFM carries it as its core. What the maintenance team learns goes back into the next design.

See Maintain →

What does maintenance cover?



Planned care of the plant and systems that keeps the facility at the standard it was handed over at: uptime protected, lifespan protected, records kept.

Does IAQ take on facilities it did not build?



Yes. Energy contracts are for facilities already running, and the tools unit is bought whether or not IAQ built the fab.



Working with IAQ

Standards, reach, and how to start.

What standards is IAQ certified to?



ISO 9001:2015, ISO 14001:2015 and ISO 45001:2018, and CIDB Grade G7 in Malaysia.

About the Group →

Where does IAQ work?



Headquartered in Shah Alam, Malaysia, with Penang, Singapore, Germany and India among its offices, and project teams across Asia and Europe. Delivering since 1995.

Global Presence →

Is the design done in BIM?



Yes. Every discipline is designed in one coordinated 3D model, so clashes are found on screen, not on site. The fab on the cover of this page is IAQ's own model.

How do I start?



Tell us what you are building, or what is running and needs to change. You have a reply within a working day.

Start a project →

Cover · One fab, every layer

Everything inside a fab, layer by layer, and which kind of work delivers each layer.

Profile: opens the business section. Website: the Services page hero. Booth screen, beside the 3D model.

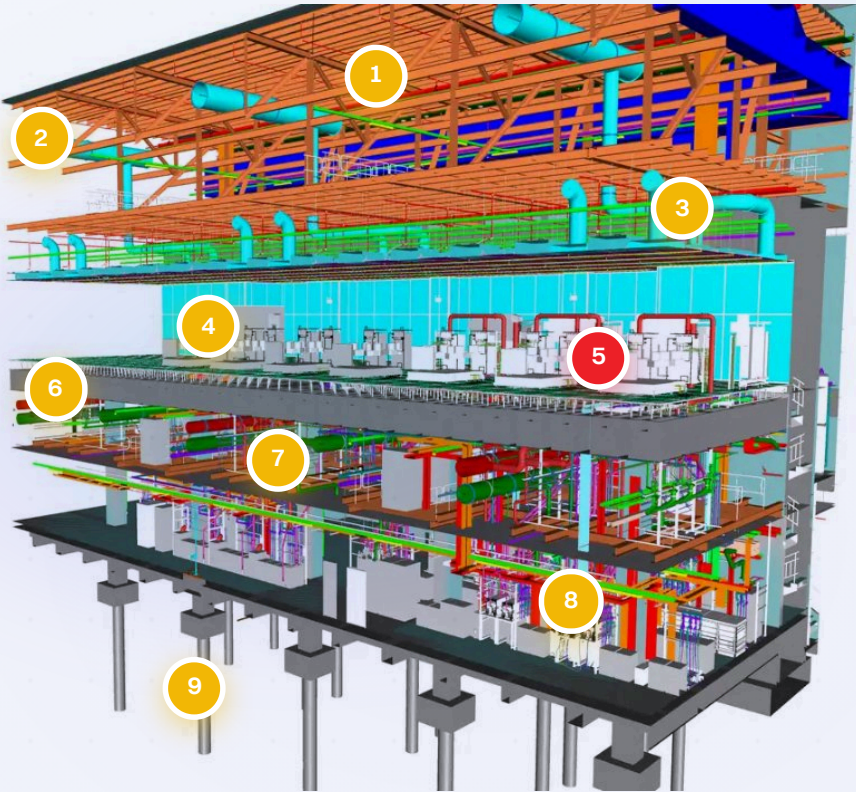
One fab, every layer: what IAQ delivers.

A section through IAQ’s own Revit coordination model. Every numbered layer is designed, procured, built and proven by the same company.

CSA (Civil, Structural and Architectural)

The building

- 1 Roof steel and structural frame
- 4 Cleanroom envelope
- 6 Waffle slab and raised floor
- 9 Piles and pile caps



MEP (Mechanical, Electrical and Plumbing)

What makes it run

- 2 Air handling and ducting
- 3 Building services, the interstitial level
- 8 Sub-fab plant and fire protection

Process utilities

What the tools run on

- 7 Process utilities in the sub-fab

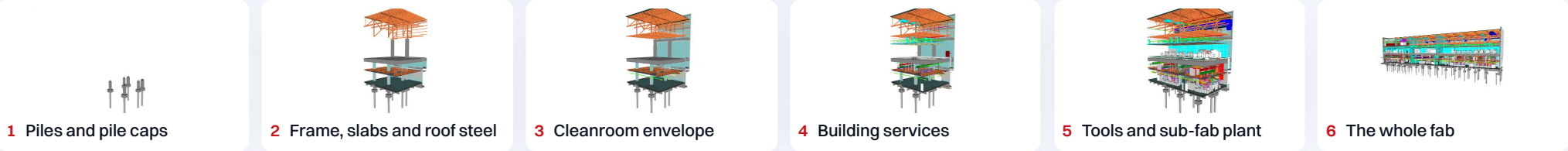
Tools hookup

Service 6, the tools themselves

- 5 Process tools, hooked up

It assembles in the order IAQ builds it

The same model, six frames of sixty one



Short forms: EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management) · BIM (Building Information Modelling)

IAQ at a glance: one group, **three business units.**

A total facility solutions provider for controlled environments. Each unit answers one question a facility owner has.



Since 1995

31 years, founded in Malaysia as a cleanroom specialist



450 people

Engineers, project teams and site crews, group wide



250+ projects

Delivered for hi-tech manufacturers and facility owners



1,050,000 m²

Of cleanroom built



7 countries

Eight offices, headquartered in Shah Alam



ISO 3

The cleanest class IAQ has built

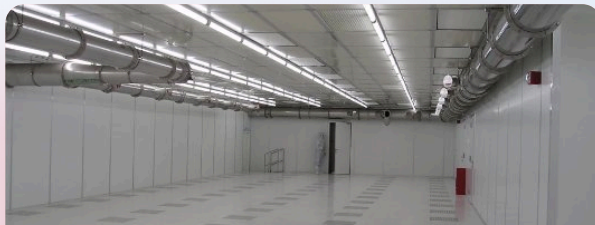


Business unit 1
EPC (Engineering, Procurement and Construction)
Builds the facility.

- Design, procurement, construction and commissioning under one contract
- Cleanrooms, dry rooms and hi-tech facilities
- CSA (Civil, Structural and Architectural), MEP (Mechanical, Electrical and Plumbing) and process utilities, in house

EPCC

EPCM



Business unit 2
Process Critical Utilities & Total Tool Installation Solutions
Re-equips a live semiconductor fab.

- Utilities: gas, chemical and slurry, water, process exhaust
- Tool installation: facilitate, rig in, hook up, commission
- Semiconductor fabs, front end to back end

Standalone

EPC or EPCM model



Business unit 3
EFM
Runs and maintains it.

- Energy Facility Management, a registered ESCO (Energy Service Company)
- IAQ funds the upgrade, paid from the savings
- Chiller plant and district cooling, run under a service level

Cooling as a Service

Energy Performance Contracting

Seven markets

 Semiconductor

 Data Centre

 EV Battery

 Photovoltaics

 District Cooling & Heating

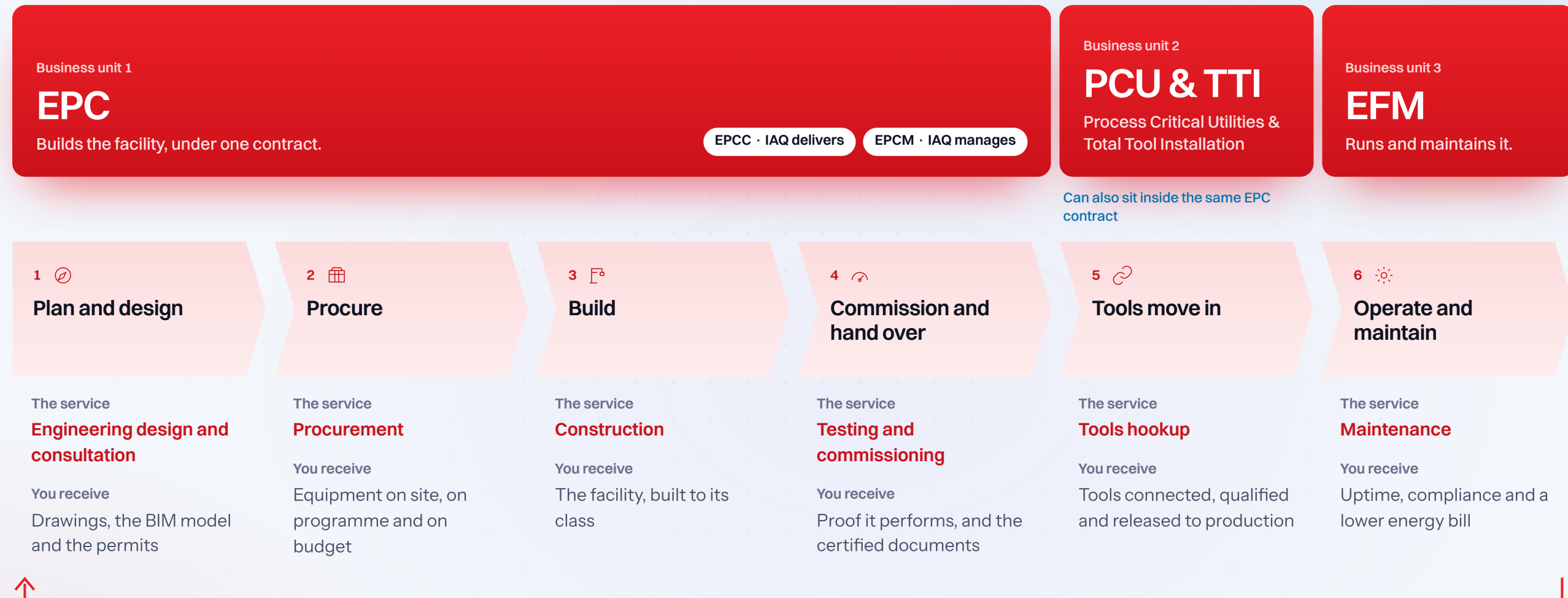
 Bio LifeScience

 Food & Beverage

Short forms: EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management)

Three units, across the life of **your facility**.

Read it as the owner does: left to right, from the first drawing to a plant in operation, and round again at the next expansion.



And round again. New tools, a new line or an energy upgrade start the next design. The same three units carry it.

Short forms: EPC (Engineering, Procurement and Construction) · EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management) · EFM (Energy Facility Management) · BIM (Building Information Modelling)



Three units across the life of a facility

● Business unit ○ Delivery model ● Service ● Work ● System ● Market

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Watch it: three short films

New to cleanrooms and chip factories? Start here. Nothing loads until you press play.

1. **What a cleanroom is** (Lighthouse Worldwide Solutions, 2:25): youtube.com/watch?v=vSS_uJlrvvk
2. **A tour of a TSMC fab** (TSMC Museum of Innovation, 7:55): youtube.com/watch?v=divOKxuYkIM
3. **Bosch opens its Penang chip plant** (Bosch Malaysia, 1:28): youtube.com/watch?v=VUG1QFOCL2E



For professionals. How the industry reads IAQ.

Engineers and buyers in hi-tech construction do not shop for a service. They ask three questions, in this order, and classify a contractor by its answers.

1 Which model?

Model

How the job is contracted. EPCC (Engineering, Procurement, Construction and Commissioning): IAQ delivers everything and answers for the result. EPCM (Engineering, Procurement and Construction Management): the owner holds the contracts and IAQ manages them. A single service on its own. Or, for energy, an ESCO (Energy Service Company) contract paid from the savings.

2 Which scope?

Service

What the contract covers. One of the six services, several, or all six. For example procurement only, design only, or design, procure and build.

3 Which work?

Work

Which kind of work the contractor does, which is how the market classifies it: CSA (Civil, Structural and Architectural), MEP (Mechanical, Electrical and Plumbing) or process utilities. Most contractors do one. IAQ does all three in house.

Everything IAQ does, and how it connects

Pick any service, unit, discipline or system. Its connections light up, and one sentence says what they mean.

Business unit	Bought as	Services it carries	Only when the client asks	Its work
EPC (Engineering, Procurement and Construction)	EPCC (Engineering, Procurement, Construction and Commissioning) or EPCM (Engineering, Procurement and Construction Management)	Engineering design and consultation, Procurement, Construction, Testing and commissioning, Maintenance and Tools hookup	None	CSA (Civil, Structural and Architectural), MEP (Mechanical, Electrical and Plumbing) and Process utilities
Process Critical Utilities & Total Tool Installation Solutions (PCU & TTI)	A standalone job, or inside an EPC (Engineering, Procurement and Construction) or EPCM (Engineering, Procurement and Construction Management) contract	Engineering design and consultation, Procurement, Construction, Testing and commissioning and Tools hookup	Maintenance	MEP (Mechanical, Electrical and Plumbing) and Process utilities
EFM (Energy Facility Management)	Cooling as a Service, Energy Performance Contracting or Build Operate Transfer	Maintenance and Testing and commissioning	Engineering design and consultation, Procurement and Construction	MEP (Mechanical, Electrical and Plumbing)

Work	The systems inside it
CSA (Civil, Structural and Architectural)	Piling and foundations, Structural frame and roof, Cleanroom envelope and Architectural finishes
MEP (Mechanical, Electrical and Plumbing)	Air conditioning and ventilation (HVAC, ACMV), fan filter units, Process cooling water (PCW), Chiller plant and district cooling, Fire protection, Electrical, high and low voltage and Plumbing and drainage
Process utilities	Clean dry air (CDA), Specialty gases and chemical delivery, Ultrapure water (UPW), Process vacuum (PV) and exhaust and Waste treatment

The same, on one chart

The chart under the cover of the Services page: which unit carries which service, the work each one does, and where that work sits in the fab.

Business unit

Unit 2 PCU & TTI

e-equips a live
emiconductor fab.

er tool set

Unit 3 EFM

uns and maintains it.

to 20 years

Always carried When you ask for it Not this unit

Work it does

- CSAMEP
- Process utilities
- Tools hookup
- MEP
- Process utilities
- Tools hookup
- MEP

Who does what, against the other kinds of contractor

The market classifies a contractor by the work it delivers. A process specialist does process only; a civil main contractor does CSA only; an M&E contractor does MEP only. IAQ delivers all five kinds, which is what "inclusive" means on this site.

Contractor	CSA	MEP	Process	Tools hookup	Energy, ESCO
IAQ Inclusive					
A process specialist Process only					
A civil main contractor CSA only					
An M&E contractor MEP only					

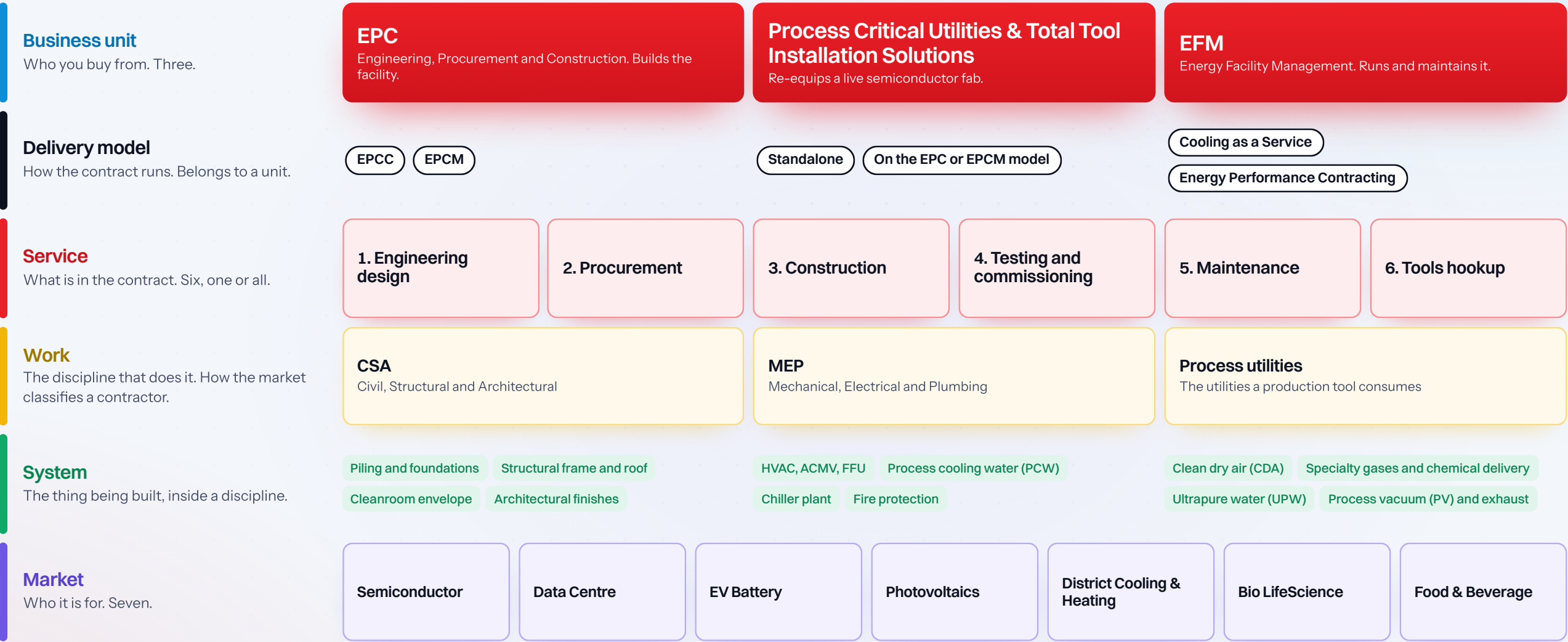
IAQ in one picture

The six kinds of thing, one colour each: unit, model, service, work, system, market.

Profile: before the scope of services. Website: Services, beside the relationship map.

IAQ in one picture: which kind of thing is which.

Six kinds of thing, one colour each, the same on every slide. Read down: who you buy from, how the contract runs, what is in it, who does it, what gets built, and for whom.



Short forms: M&E (mechanical and electrical)

Three words that mean two things.

Found by cross-checking IAQ's own documents against the 17 September review and the site. The colour says which kind of thing each meaning is.

EPC

One word, used for three different kinds of thing. This is where most of the confusion starts.

Business unit

EPC (Engineering, Procurement and Construction), **unit 1**
The unit that builds the facility.

Delivery model

The EPC model: EPCC (Engineering, Procurement, Construction and Commissioning), **EPCM** (Engineering, Procurement and Construction Management)
How a contract runs. The hookup unit works on it too.

A contract

Energy Performance Contracting
EFM (Energy Facility Management)'s contract. Always written in full, never EPC.

Say it this way

Say "the EPC unit" or "the EPC model". Never shorten the energy contract.

Hookup

Unit 2 went by two names, and one of them borrowed the name of a service.

Business unit

Process Critical Utilities & Total Tool Installation Solutions

Unit 2, IAQ's own written name. "Total Tools Hookup Solution" is retired.

Service

Tools hookup, service 6

The work of connecting a tool. The only thing "hookup" names now.

Say it this way

Name the unit by what it sells. Keep "hookup" for the work.

PCW, CDA, UPW

Systems turn up in two places, so they get listed as services by mistake.

System

Built with the facility

PCW (process cooling water) is mechanical work. CDA (clean dry air) and UPW (ultrapure water) are process work.

Service

Their last metres are hookup

From the main to the tool, service 6 connects them.

Say it this way

A system is never a service. It appears twice because two crews touch it.

Six services. Buy one, or all of them.

IAQ calls the six its total solution: the scope written into the contract, in the order a project runs. Under it, the business unit that carries each one.

	Before site		On site		In operation	
The service	1 Engineering design and consultation	2 Procurement	3 Construction	4 Testing and commissioning	5 Maintenance	6 Tools hookup
The scope	- Feasibility and master planning - Concept to detailed design, CSA (Civil, Structural and Architectural), MEP and process - BIM coordination - Permitting and authority liaison	- Supplier prequalification - Tenders, estimates and cost analysis - Long-lead equipment tracked	- Site and construction management - Safety, quality assurance and control - Programme and cost control	- Cleanroom classification testing - System performance verification - As-built and handover documents	- Planned preventive maintenance - Breakdown response - Optimisation and retrofits	- Tool move-in and placement - Utilities tied in to each tool - Qualified and released to production
Carried by EPC						
PCU & TTI					when asked	
EFM	when asked	when asked	when asked			



Every discipline
Civil, structural and architectural, MEP (Mechanical, Electrical and Plumbing), and process utilities, in house.



Any combination
One service, several, or all six. The model that governs it stays the same.



Two delivery models
EPCC, IAQ delivers the whole project. EPCM, IAQ manages it for the owner.

Short forms: BIM (Building Information Modelling) · EPC (Engineering, Procurement and Construction) · PCU & TTI (Process Critical Utilities & Total Tool Installation) · EFM (Energy Facility Management) · EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management)

Four ways to contract IAQ

How the work is bought: EPCC, EPCM, one service alone, or an energy contract.

Profile: after scope of services. Website: the EPC (Engineering, Procurement and Construction) page.

Four ways to contract IAQ.

The model governs the work. The scope, one service or all six, sits inside it. In EPCC (Engineering, Procurement, Construction and Commissioning), IAQ does the work. In EPCM (Engineering, Procurement and Construction Management), IAQ manages the work for the owner.



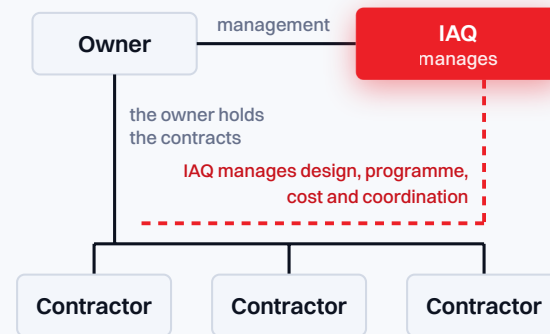
EPCC

IAQ delivers the complete project, commissioning included, and answers for how the system performs.

Contracts held by	IAQ
Answers for the result	IAQ
Typical term	12 to 24 months

Best when

You want one accountable party.



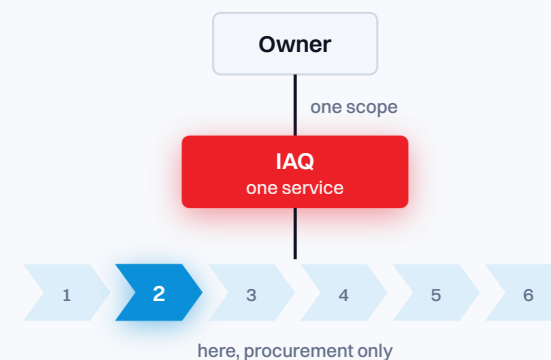
EPCM

The owner holds the construction contracts. IAQ manages design, contractors, programme and cost.

Contracts held by	The owner
Answers for the result	IAQ manages, the contractors deliver
Typical term	12 to 24 months

Best when

You want direct control of the contracts, often on the largest programmes.



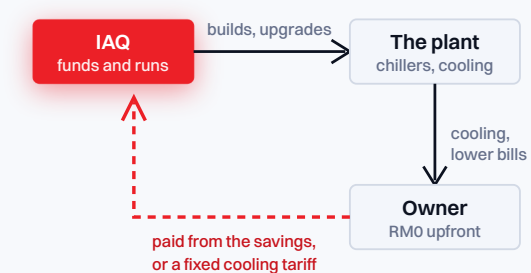
A single service

Any one of the six on its own: procurement only, design only, hookup only. The same model governs it.

Contracts held by	IAQ, for that scope
Answers for the result	IAQ, for that scope
Typical term	Set by the scope

Best when

You already hold the rest of the scope.



ESCO (Energy Service Company) contract

EFM (Energy Facility Management) funds and runs the energy upgrade. Cooling as a Service, or Energy Performance Contracting.

Funded by	IAQ
Answers for the result	IAQ, for the savings
Typical term	5 to 20 years

Best when

You want lower energy cost with no capital spent.

Short forms: CSA (Civil, Structural and Architectural) · MEP (Mechanical, Electrical and Plumbing)

The work: three disciplines

The work a contractor is classified by, and the systems inside it.

Profile: scope of work. Website: Services, the glossary's place.

Three disciplines. One contractor.

The market classifies a contractor by the work it delivers. A specialist carries one discipline. IAQ carries all three. The drawings are IAQ's own Revit model.

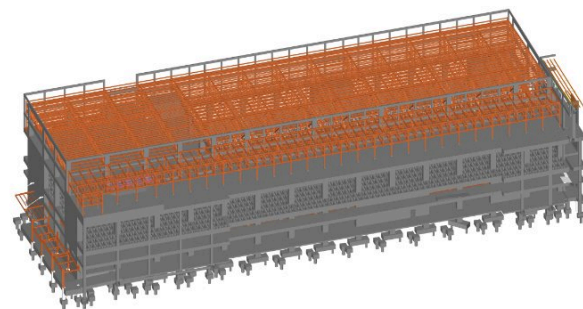


CSA

Civil, Structural and Architectural

The building itself.

- Piling and foundations
- Structural frame and roof
- Cleanroom envelope: walls, ceiling grid, raised floor
- Architectural finishes

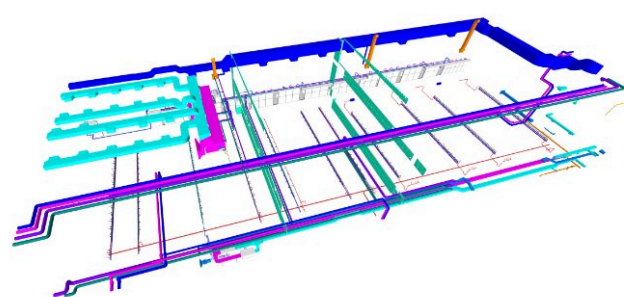


MEP

Mechanical, Electrical and Plumbing

What makes the building run.

- HVAC, ACMV and fan filter units
- Process cooling water (PCW)
- Chiller plant and district cooling
- Fire protection
- Electrical, high and low voltage
- Plumbing and drainage

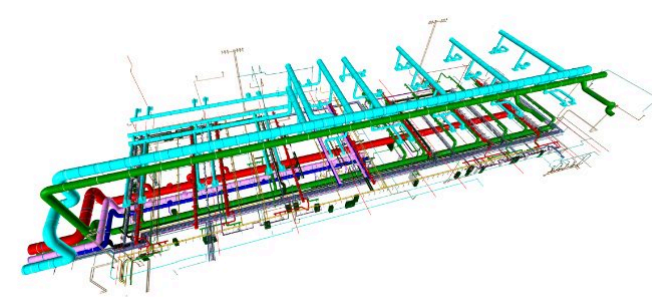


Process utilities

The utilities a production tool consumes

What the tool runs on.

- Clean dry air (CDA)
- Specialty gases and chemical delivery
- Ultrapure water (UPW)
- Process vacuum (PV) and exhaust
- Waste treatment



IAQ

Inclusive: civil to process, under one roof

A civil contractor

CSA (Civil, Structural and Architectural) only

An M&E contractor

MEP (Mechanical, Electrical and Plumbing) only

A process specialist

Process only

Short forms: HVAC (heating, ventilation and air conditioning) · ACMV (air conditioning and mechanical ventilation) · M&E (mechanical and electrical)



The work: three disciplines

● Business unit ○ Delivery model ● Service ● Work ● System ● Market

7 / 11

EPC: one contract, from first drawing to final handover.

Engineering, Procurement and Construction. IAQ designs the system, procures it, builds it and proves it operates as intended.

1

point of contact for the whole facility

12 to 24

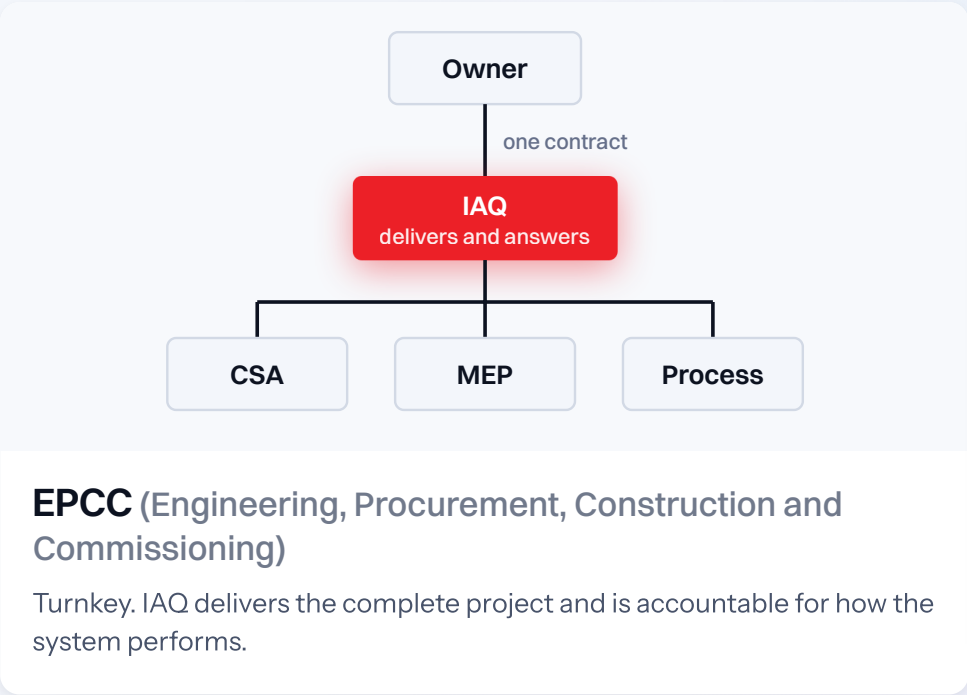
months, a typical EPC programme

6

services, from design to tools hookup

3

disciplines in house: CSA, MEP, process



At a glance	EPCC	EPCM
Who holds the construction contracts	IAQ	The owner
Who answers for the system performing	IAQ	IAQ manages, the owner's contractors deliver
Commissioning and start-up	Included	Coordinated by IAQ
Typical programme	12 to 24 months	12 to 24 months

Short forms: CSA (Civil, Structural and Architectural) · MEP (Mechanical, Electrical and Plumbing)

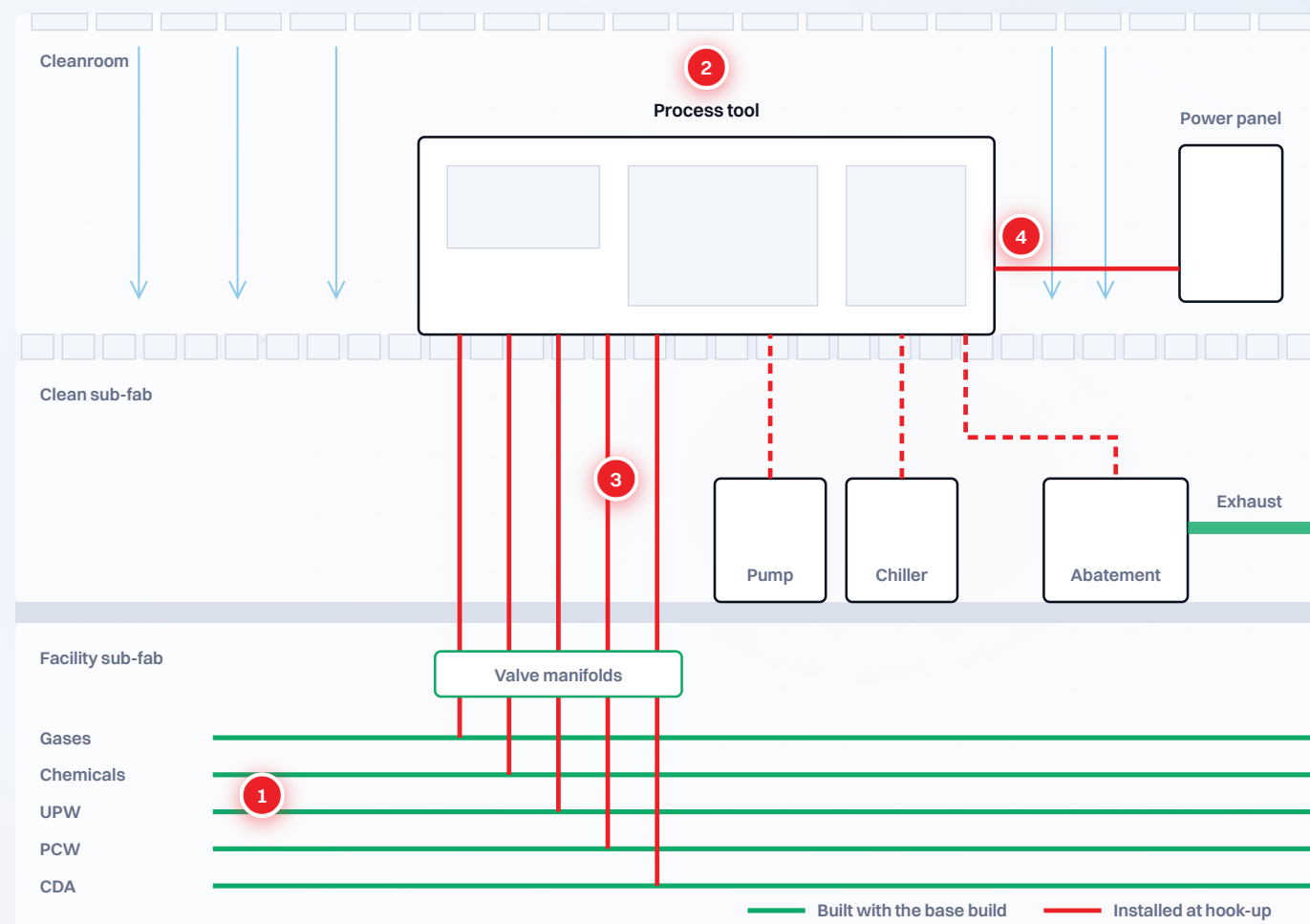
From a delivered tool to a producing tool.

Two halves, one unit, for semiconductor fabs only. Process Critical Utilities builds the utilities with the base build. Total Tool Installation then connects each tool to them.

Process Critical Utilities

IAQ's five solution groups

- **Water**
UPW (ultrapure water) plant and distribution, process drains, waste water treatment, reclaim
- **Chemical and slurry**
Bulk chemical, blending, dispense, storage tanks
- **Gas**
Bulk and specialty gas, abatement, bunkers, liquid dispense
- **Process exhaust**
General, acid, caustic, solvent and calamity
- **Space management**
The sub-fab planned in 3D BIM (Building Information Modelling), off-site modules



Total Tool Installation

IAQ's four phases, marked on the section

- 1 Facilitization**
Before the tool arrives: drops, power, structure and room built to the maker's requirements.
- 2 Rig-in and move-in**
Uncrated in an airlock, moved on an agreed route, set, levelled and isolated.
- 3 Hook-up**
Each utility tapped from its valve manifold and connected into the tool.
- 4 Commissioning**
Leak tested, purged, balanced, trip tested, then handed to operations.

How it is bought

On its own, even in a fab IAQ did not build, or inside an EPCC (Engineering, Procurement, Construction and Commissioning) or EPCM (Engineering, Procurement and Construction Management) contract.

Short forms: PCW (process cooling water) · CDA (clean dry air)

EFM: lower energy bills, funded by IAQ.

Energy Facility Management, a registered ESCO (Energy Service Company) under Suruhanjaya Tenaga. IAQ funds, builds and runs the upgrade.

Two ways to pay for it

10 to 20 years

Cooling as a Service

IAQ finances, builds or rehabilitates, and operates the chiller plant. A fixed tariff for the cooling used, under a service level agreement. The plant is handed over at the end.

5 to 10 years

Energy Performance Contracting

IAQ funds and delivers the upgrades. Every payment comes from the savings actually measured. After the contract, the owner keeps all of them.



The four problems it solves

High electricity bills

Inefficient cooling, driving cost up year after year.

Rising carbon emissions

Compliance risk against regulation and targets.

Ageing infrastructure

Chiller plant past its best, and a capital bill to replace it.

Operational disruption

Downtime from old systems and poor maintenance.

Proof, by market and by the work delivered.

The 18 published references of 250+ delivered projects. Each light is one project; client names are withheld.



Malaysia	14
Singapore	1
China	1
Europe	2

The full record is 250+ projects. This slide grows with the approved project list: every project added to the registry lights here on its own.

Watch it, by business unit

Nine short tutorials. Each one explains a term used on this page.

EPC (Engineering, Procurement and Construction)

PCU & TTI (Process Critical Utilities & Total Tool Installation)

EFM (Energy Facility Management)

- 1. **EPC and EPCM, explained** (Project Business, 6:17): [youtube.com/watch?v=QdgS9MBDZSA](https://www.youtube.com/watch?v=QdgS9MBDZSA)
- 2. **What the ISO cleanroom classes mean** (Sure-BioChem Laboratories, 3:20): [youtube.com/watch?v=IENbOfC4df8](https://www.youtube.com/watch?v=IENbOfC4df8)
- 3. **Inside the Morrow Batteries dry room** (Munters, 3:17): [youtube.com/watch?v=dGd7hXcx_7w](https://www.youtube.com/watch?v=dGd7hXcx_7w)
- 4. **How TSMC builds a 19 billion dollar fab** (Asianometry, 12:50): [youtube.com/watch?v=785Uzi1mGAA](https://www.youtube.com/watch?v=785Uzi1mGAA)
- 5. **Inside an Intel sub fab** (Intel Newsroom, 3:02): [youtube.com/watch?v=8jpRADL3BGM](https://www.youtube.com/watch?v=8jpRADL3BGM)
- 6. **Ultrapure water for chip making** (Asianometry, 12:50): [youtube.com/watch?v=C3RzODSR3gk](https://www.youtube.com/watch?v=C3RzODSR3gk)
- 7. **District cooling in two minutes** (International District Energy Association, 1:46): [youtube.com/watch?v=tEh16-NusuQ](https://www.youtube.com/watch?v=tEh16-NusuQ)
- 8. **Cooling as a Service** (BASE Foundation, 2:31): [youtube.com/watch?v=o2711xsqiBw](https://www.youtube.com/watch?v=o2711xsqiBw)
- 9. **Energy Performance Contracting** (Codema, Dublin's energy agency, 6:33): [youtube.com/watch?v=kWF0ywtEvhY](https://www.youtube.com/watch?v=kWF0ywtEvhY)

Six buyer requests, each read as a path through the layers

“A new fab, one party accountable for the result.”

Unit	EPC
Model	EPCC
Scope	DesignProcureConstructCommissionHook up
Work	CSAMEPProcess utilities
System	The whole facility

“We hold the contracts. Manage the programme for us.”

Unit	EPC
Model	EPCM
Scope	DesignProcureConstructCommission
Work	CSAMEPProcess utilities
System	The whole facility, through the owner’s contractors

“Buy the process cooling water package for us, that one package.”

Unit	EPC
Model	EPC, one service
Scope	Procure
Work	MEP
System	Process cooling water (PCW)

“Design, procure and build the PCW loop.”

Unit	EPC
Model	EPC, three services
Scope	DesignProcureConstruct
Work	MEP
System	Process cooling water (PCW)

“Forty tools land in March. The fab is live.”

Unit	PCU & TTI (Process Critical Utilities & Total Tool Installation)				
Model	Standalone, or under EPC				
Scope	Design	Procure	Construct	Commission	Hook up
Work	MEP	Process utilities			
System	Gases, chemicals, UPW (ultrapure water), exhaust and power, to each tool				

“Cut the cooling bill, with IAQ funding the work.”

Unit	EFM	
Model	Cooling as a Service	
Scope	Maintain	Commission
Work	MEP	
System	Chiller plant, run by IAQ under a tariff	

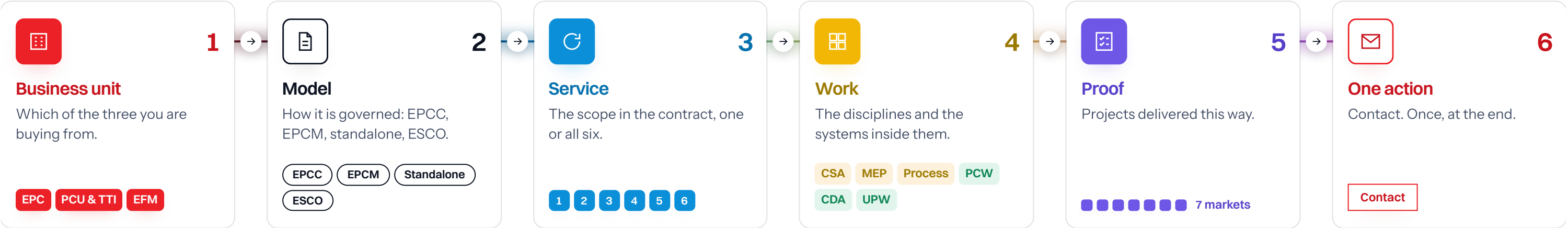
3

On the website. How the site shows it, page by page.

The same order everywhere: business unit first, then how it is bought, what is in it, who does it, the proof, and one action. What is live now and what is proposed.

The order a visitor reads in

Every page reads in this order, and ends on one action.



One ask from the review, read in order: “Buy the process cooling water package for us, that one package”



The Services menu, live now

Three columns: the picture, the three units as cards, the six services as one list. Point at a unit to see what it carries.

IAQ

AboutServicesMarketsProjectsNewsCareersLive on the site

Services

The six services of the delivery cycle, and the three business units the work is contracted through.

Open the page →

Business units

EPC

Engineering, Procurement and Construction

Builds the facility, under one contract.

EPCC EPCM

Process Critical Utilities & Total Tool Installation Solutions

Re-equips a live semiconductor fab.

Process Critical UtilitiesTotal Tool Installation

EFM

Energy Facility Management

Runs and maintains it.

Cooling as a ServiceEnergy Performance Contracting

Six services

1Engineering Design

2Procurement

3Construction

4Testing & Commissioning

5Maintenance

6Tools Hookup

1 Three columns: the picture, the units, the services.

The three business units are three cards, because a buyer chooses the unit first.

2 Every section inside a unit is a link.

Its own pages, and its contract models, which open “Two ways to buy it” on the unit’s page.

3 The six services, one numbered list.

Point at a unit and the list lights what it carries, and marks what it carries only when asked.

The Services page, top to bottom

Seven bands in the reading order, as the page stands on 26 September 2026. Point at a band to see its job, click it to see its slide.

iaqtechnology.com.my/services

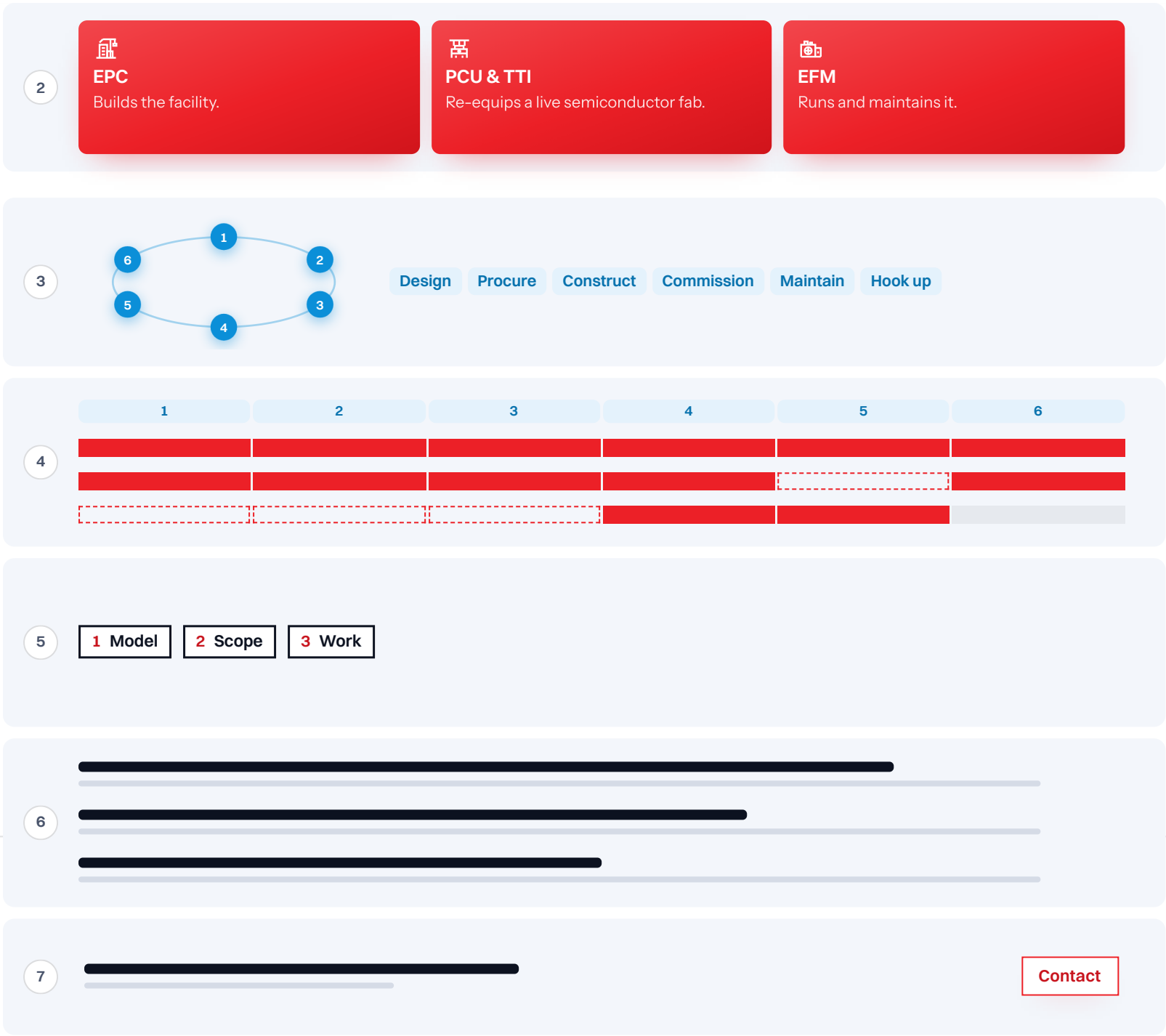
1

Live built today Proposed to build

The banner: the IAQ facility map Live

IAQ’s own Revit model on the left (drag to turn, zoom, a tour you play), and on the right the map of 6 services, 3 business units, 3 kinds of work and 15 systems. Pick anything and the model builds to that moment and lights the layers it touches; the reading of the relation sits under the model, at one fixed height.

Business unit Model Service Work



The three business units, one card each ● Live

White cards with the unit’s photograph, its tag, its mark beside the name and its one line; then when to call it and how it is bought. Read more opens the three in detail as one bar across the cards. Each card opens its page.

■ Business unit

Six services, one accountable team ● Live

The horizontal serpentine: 1 to 6 left to right, the U-turn, the red return from Hookup into the next Design. It steps on its own until touched. Each stage opens its page.

■ Service

Six services, three units, four kinds of work, in one view ● Live

The one-view figure: the six services across, the three units as bars over the services they carry (solid always, dashed when asked), the kinds of work with who does each.

□ Model ■ Service ■ Work

What to settle before you ask for a quote ● Live

Three questions (model, scope, work) and six typical requests read as a path through the layers.

□ Model

Questions clients ask, answered ● Live

The questions in five groups (choosing a model, time and cost, a live fab, after handover, working with IAQ), each answer one paragraph with the page that says more. Codex part 1 holds the same set.

One action: contact ● Live

The closing band. The only call to action on the page.

□ One action

A business unit page, part by part

Eight parts, the same on all three unit pages, so they compare line for line; EFM carries its seven services where the models stand. EPC shown.

iaqtechnology.com.my/services/epc-construction

EPC

Engineering, Procurement and Construction

EPCCEPCM

One contract for a hi-tech facility, from the first drawing to handover.

EPCC

EPCC

EPCM

EPCM

CSA

MEP

Process

1

2

3

4

5

6

Semiconductor

Data Centre

EV Battery

Contact

- 1

Name and models
EPC, its facts and its photograph. Bought as EPCC or EPCM.
- 2

What it is
In plain terms, the unit’s own words; on EFM, the problems it solves under their own heading.
- 3

The models
One card each: who holds the contracts, who answers for the result, the term. On EFM, the seven energy management services stand here instead.
- 4

The work
CSA, MEP, process, and the systems under each; on EFM, district cooling in brief first.
- 5

The cycle
The six stages beside the unit’s drawing; each stage names what IAQ carries there.
- 6

Why owners choose it
Six claims, one line each; a click opens the sentence behind it and never closes it.
- 7

Proof
Projects tagged with this model, by scope and location.
- 8

One action
Contact, once.

Where every picture goes

The twelve slides, and where each one is placed. Click a slide to open it full size.

One fab, every layer: what IAQ delivers.

A section through IAQ's own Revit coordination model. Every numbered layer is designed, procured, built and proven by the same company.

CSA (Civil, Structural and Architectural)
The building

1 Roof steel and structural frame

2 Cleanroom envelope

3 Wafer slab and raised floor

4 Piles and pile caps

MEP (Mechanical, Electrical and Plumbing)
What makes it run

1 Air handling and ducting

2 Building services, the interstitial level

3 Sub-fab plant and fire protection

Process utilities
What the tools run on

1 Process utilities in the sub-fab

Tools hookup
Service 8, the tools themselves

1 Process tools, hooked up

It assembles in the order IAQ builds it

The same model, six frames of sixty one

1 Piles and pile caps

2 Frame, slab and roof steel

3 Cleanroom envelope

4 Building services

5 Tools and sub-fab plant

6 The whole fab

IAQ One fab, every layer

Business unit

Delivery model

Service

Work

System

Market

Cover

Cover · One fab, every layer

On the site

The Services page hero.

In the profile

Opens the business section.

Booth screen, beside the 3D model.

IAQ at a glance: one group, three business units.

A total facility solutions provider for controlled environments. Each unit answers one question a facility owner has.

Since 1995

21 years, founded in Malaysia as a cleanroom specialist

450 people

Engineers, project teams and site crews, group wide

250+ projects

Delivered for hi-tech manufacturers and facility owners

1,050,000 m²

Of cleanroom built

7 countries

Eight offices, headquartered in Shah Alam

ISO 3

The cleanest class IAQ has built

Business unit 1
EPC (Engineering, Procurement and Construction)
Builds the facility.

Design, procurement, construction and commissioning under one contract

Cleanrooms, dry rooms and hi-tech facilities

CSA (Civil, Structural and Architectural), MEP (Mechanical, Electrical and Plumbing) and process utilities, in-house

Business unit 2
Process Critical Utilities & Total Tool Installation Solutions
Re-equips a live semiconductor fab.

Utilities: gas, chemical and slurry, water, process exhaust

Tool installation, facilities, rig in, hook up, commission

Semiconductor fabs, front end to back end

Business unit 3
EFM
Runs and maintains it.

Energy Facility Management, a registered ESCO (Energy Service Company)

IAQ funds the upgrade, paid from the savings

Chiller plant and district cooling, run under a service level

Services markets

Semiconductor

Data Centre

EV Battery

Photovoltaics

District Cooling & Heating

Bio LifeScience

Food & Beverage

IAQ at a glance

Business unit

Delivery model

Service

Work

System

Market

1 / 11

IAQ at a glance

On the site

About.

In the profile

The opening spread.

Three units, across the life of your facility.

Read it as the owner does: left to right, from the first drawing to a plant in operation, and round again at the next expansion.

Business unit 1
EPC
Builds the facility, under one contract.

Business unit 2
PCU & TTI
Process Critical Utilities & Total Tool Installation

Business unit 3
EFM
Runs and maintains it.

Can use or reuse the same EPC (Engineering, Procurement and Construction) contract

1 Plan and design

2 Procure

3 Build

4 Commission and hand over

5 Tools move in

6 Operate and maintain

The service
Engineering design and consultation

You receive
Drawings, the BIM model and the permits

The service
Procurement

You receive
Equipment on site, on programme and on budget

The service
Construction

You receive
The facility, built to its class

The service
Testing and commissioning

You receive
Proof it performs, and the certified documents

The service
Tools hookup

You receive
Tools connected, qualified and released to production

The service
Maintenance

You receive
Uptime, compliance and a lower energy bill

And round again. New tools, a new line or an energy upgrade start the next design. The same three units carry it.

Short forms: EPCO (Engineering, Procurement, Construction and Commissioning); EPCM (Engineering, Procurement and Construction Management); EFM (Energy Facility Management); BIM (Building Information Modeling)

IAQ Three units across the life of a facility

Business unit

Delivery model

Service

Work

System

Market

2 / 11

Three units across the life of a facility

On the site

Services, first screen.

In the profile

The page on the three business units.

Booth screen.

IAQ in one picture: which kind of thing is which.

Six kinds of thing, one colour each, the same on every slide. Read down: who you buy from, how the contract runs, what is in it, who does it, what gets built, and for whom.

Business unit
Who you buy from. Three.

EPC
Engineering, Procurement and Construction. Builds the facility.

Process Critical Utilities & Total Tool Installation Solutions
Re-equips a live semiconductor fab.

EFM
Energy Facility Management. Runs and maintains it.

Delivery model
How the contract runs. Belongs to a unit.

EPCO

EPCM

Stand-alone

On the EPC or EPCM model

Cooling as a Service

Energy Performance Contracting

Service
What is in the contract. Six, one or all.

1. Engineering design

2. Procurement

3. Construction

4. Testing and commissioning

5. Maintenance

6. Tools hookup

Work
The discipline that does it. How the market classifies a contractor.

CSA
Civil, Structural and Architectural

MEP
Mechanical, Electrical and Plumbing

Process utilities
The utilities a production-fab consumes

System
The thing being built, inside a discipline.

Wing and foundations

Structural frame and roof

HVAC, ACMV, RTU

Process cooling water (PCW)

Clean dry air (CDA)

Specialty gases and chemical delivery

Cleanroom envelope

Architectural finishes

Chiller plant

Fire protection

Ultra-pure water (UPW)

Process vacuum (PV) and exhaust

Market
Who it is for. Seven.

Semiconductor

Data Centre

EV Battery

Photovoltaics

District Cooling & Heating

Bio LifeScience

Food & Beverage

Short forms: EPCO (Engineering, Procurement, Construction and Commissioning); EPCM (Engineering, Procurement and Construction Management); HVAC (heating, ventilation and air conditioning); ACMV (air conditioning and mechanical ventilation); RTU (fan filter unit)

IAQ in one picture

Business unit

Delivery model

Service

Work

System

Market

3 / 11

IAQ in one picture

On the site

Services, beside the relationship map.

In the profile

Before the scope of services.

Three words that mean two things.

Found by cross-checking IAQ's own documents against the 17 September review and the site. The colour says which kind of thing each meaning is.

EPC

One word, used for three different kinds of thing. This is where most of the confusion starts.

Business unit
EPC (Engineering, Procurement and Construction), unit 1
The unit that builds the facility

The EPC model: EPCO (Engineering, Procurement, Construction and Commissioning), EPCM (Engineering, Procurement and Construction Management)
How a contract runs. The hookup unit works on it too.

Delivery model
Energy Performance Contracting
EFM (Energy Facility Management) is contract. Manage written in full, never EPC.

Hookup

Unit 2 went by two names, and one of them borrowed the name of a service.

Business unit
Process Critical Utilities & Total Tool Installation Solutions
Unit 2, IAQ's own written name. "Total Tools Hookup Solution" is written.

Service
Tools hookup, service 8
The work of connecting a tool. The only thing "hookup" names now.

PCW, CDA, UPW

Systems turn up in two places, so they get listed as services by mistake.

System
Built with the facility
PCW (process cooling water) is mechanical work. CDA (clean dry air) and UPW (ultra-pure water) are process work.

Service
Their last metres are hookup
From the main to the tool, service 8 connects them.

Say it this way
Name the unit by what it sells. Keep "hookup" for the work.

Say it this way
A system is never a service. It appears twice because two screens found it.

IAQ The words that got mixed up

Business unit

Delivery model

Service

Work

System

Market

4 / 11

The words that get mixed up

On the site

Not on the site.

In the profile

Not in the profile.

Internal first: the team, the profile writers, the booth crew. Then the glossary's place on the site.

Six services. Buy one, or all of them.

IAQ calls the six its total solution: the scope written into the contract, in the order a project runs. Under it, the business unit that carries each one.

The service

Before site

On site

In operation

1. Engineering design and consultation

2. Procurement

3. Construction

4. Testing and commissioning

5. Maintenance

6. Tools hookup

The scope

Feasibility and master planning

Concept to detailed design, CSA (Civil, Structural and Architectural), MEP and process

BIM coordination

Permitting and authority liaison

Supplier pre-qualification

Tenders, estimates and cost analysis

Long-lead equipment tracked

Site and construction management

Safety, quality assurance and control

Programme and cost control

Cleanroom classification testing

System performance verification

As-built and handover documents

Planned preventive maintenance

Breakdown response

Optimisation and retrofits

Tool move-in and placement

Utilities tied in to each tool

Qualified and released to production

Controlled by

EPC

PCU & TTI

EFM

Every discipline

Civil, structural and architectural, MEP (Mechanical, Electrical and Plumbing), and process utilities, in-house

Any combination

One service, several, or all six. The model that governs it shapes the same

Two delivery models

EPCO, IAQ delivers the whole project. EPCM, IAQ manages it for the owner

Short forms: BIM (Building Information Modeling); EPC (Engineering, Procurement and Construction); PCU & TTI (Process Critical Utilities & Total Tool Installation); EFM (Energy Facility Management); EPCO (Engineering, Procurement, Construction and Commissioning); EPCM (Engineering, Procurement and Construction Management)

IAQ Six services

Business unit

Delivery model

Service

Work

System

Market

5 / 11

Six services

On the site

Services, under the units.

In the profile

Scope of services.

Four ways to contract IAQ.

The model governs the work. The scope, one service or all six, sits inside it. In EPCC (Engineering, Procurement, Construction and Commissioning), IAQ does the work. In EPCM (Engineering, Procurement and Construction Management), IAQ manages the work for the owner.



Short forms: CSA (Civil, Structural and Architectural); MEP (Mechanical, Electrical and Plumbing)

IAQ Four ways to contract IAQ

Four ways to contract IAQ

- On the site The EPC page.
- In the profile After scope of services.

Three disciplines. One contractor.

The market classifies a contractor by the work it delivers. A specialist carries one discipline. IAQ carries all three. The drawings are IAQ's own Revit model.



Short forms: HVAC (heating, ventilation and air conditioning); AC/MV (air conditioning and mechanical ventilation); M&E (mechanical and electrical)

IAQ The work: three disciplines

The work: three disciplines

- On the site Services, the glossary's place.
- In the profile Scope of work.

EPC: one contract, from first drawing to final handover.

Engineering, Procurement and Construction. IAQ designs the system, procures it, builds it and proves it operates as intended.



Short forms: CSA (Civil, Structural and Architectural); MEP (Mechanical, Electrical and Plumbing)

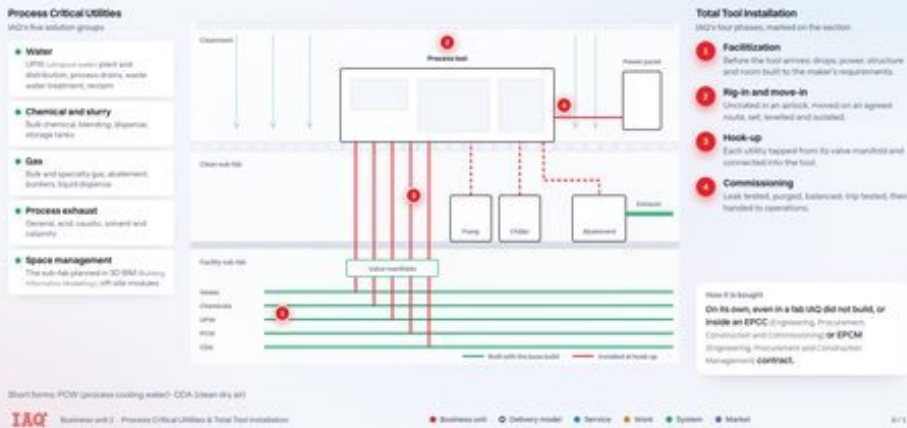
IAQ Business unit 1 - EPC

Business unit 1 · EPC

- On the site Not on the site.
- In the profile Extension page 1, after the page on the three business units.

From a delivered tool to a producing tool.

Two halves, one unit, for semiconductor fabs only. Process Critical Utilities builds the utilities with the base build. Total Tool Installation then connects each tool to them.



Short forms: PCW (process cooling water); CDA (clean dry air)

IAQ Business unit 2 - Process Critical Utilities & Total Tool Installation

Business unit 2 · Process Critical Utilities & Total Tool Installation

- On the site Not on the site.
- In the profile Extension page 2.

Booth screen beside the 3D model.

EFM: lower energy bills, with no upfront cost.

Energy Facility Management, a registered ESCO (Energy Service Company) under Suruhanjaya Tenaga. IAQ funds, builds and runs the upgrade.



Short forms: EFM (Energy Facility Management)

IAQ Business unit 3 - EFM

Business unit 3 · EFM

- On the site Not on the site.
- In the profile Extension page 3.

Proof, by market and by the work delivered.

The 18 published references of 250+ delivered projects. Each light is one project; client names are withheld.



Short forms: EFM (Energy Facility Management)

IAQ Proof, by market and by work delivered

Proof, by market and by work delivered

- On the site Projects.
- In the profile Opens the project references.

Interactive moments to build

Four interactive pieces that show how the parts connect. Two are live on the site, two are proposed.

Systems

Process

MEP

CSA

The fab build-up

Home page, in 3D. Press play and the fab assembles layer by layer, each layer named by its work.

Proposed

 Asked for on 17 Sep

123

456

The scope lights

Each unit page. The six services, the unit’s own lit, the ones carried on request ringed.

Live

EPC

EPCC

1

2

MEP

EFM

EPCM

3

4

PCW

The relationship map

Services page. Pick a unit, and its models, services, work and systems light up.

Live

1Facilitize

2Rig in

3Hook up

4Commission

The hook-up section

Total Tool Installation page. IAQ’s four phases, from facilitization to commissioning.

Live

One name each, one goal per page

The names

~~EPC & Construction~~

→

EPC, Engineering, Procurement and Construction

EPC already includes construction. Nabilah, 17 Sep.

~~Total Tools Hookup Solution~~

→

Process Critical Utilities & Total Tool Installation Solutions

IAQ’s own name in the Discovery answers, the website copy sheet, the questionnaire and Nabilah’s 18 Sep message. “Hookup” stays the name of service 6, so one word no longer names two things.

~~Energy Management~~

→

EFM, Energy Facility Management

Nabilah’s 18 Sep name for unit 3. The Discovery answers say Energy Management Solutions; the questionnaire confirms it covers what EFM provides.

~~EPC, for the energy contract~~

→

Energy Performance Contracting, always spelled out

EPC means Engineering, Procurement and Construction everywhere else.

The goals	
Home	Understand what IAQ is, then choose a business unit.
Services	Choose the business unit.
A unit page	Contact, knowing the model, scope and work.
Markets	Find your own sector, then its proof.
Projects	Filter the record to work like yours.
About	Trust: the record, the certifications, the people.
News	Read the latest.
Careers	Apply.

In the company profile

IAQ, 18 September: the three unit pages go into the profile straight after the page on the three business units. Every picture above has its place.

C1	Opening The cover, then who IAQ is in one view.
2	The page on the three business units The three units along the life of the owner’s facility.
8910	Directly after it, the extension pages IAQ asked for on 18 Sep EPC (Engineering, Procurement and Construction), then Process Critical Utilities & Total Tool Installation, then Energy Facility Management.
3567	Scope of services The one picture, the six services, the four ways to contract, the three disciplines.
11	Opening the project references Proof by market and by work delivered.

Short forms: EPCC (Engineering, Procurement, Construction and Commissioning) · EPCM (Engineering, Procurement and Construction Management) · ESCO (Energy Service Company) · CSA (Civil, Structural and Architectural) · MEP (Mechanical, Electrical and Plumbing) · CDA (clean dry air) · UPW (ultrapure water)

IAQ’s own documents, checked line by line.

The Discovery answers, the three business unit questionnaires, the website copy sheet, the Utility Solutions deck, the moodboard, the annotated feedback deck, the 17 September review and Nabilah’s 18 September message, read against the site. Where they disagree, IAQ’s written word wins.

5

Fixed on the site

6

Already agreed

3

IAQ's papers disagree

Fixed on the site The name of business unit 2	IAQ’s documents Discovery answers A2.1 Process Critical Utilities & Total Tool Installation Solutions Website copy sheet the same Nabilah, 18 Sep message Process Critical Utilities and Total Tool Installation Solutions 17 Sep review flags two names in use
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Already agreed	IAQ's documents		The site said	Now
	Unit 2 questionnaire	Standalone, but using the EPC (Engineering, Procurement and Construction) or EPCM (Engineering, Procurement and Construction Management) model	Standalone, or on the EPC or EPCM model	Unchanged.
	Utility Solutions deck	A specialist EPCM partner		
IAQ's papers disagree	IAQ's documents		The site said	Now
	Discovery answers A2.1	Three business models	Three	Three. The copy sheet's "four" reads as a leftover.
	Website copy sheet	Headline "Four Ways We Deliver Certainty", body "three core business models", notes "4 clickable model cards"		
Already agreed	IAQ's documents		The site said	Now
	Discovery answers A1.3	Not core services: the Total Solutions, six, Tools Hookup added	Six services	Six services, subtitled IAQ's total solution.
	17 Sep review	The six services are the scope		
Already agreed	IAQ's documents		The site said	Now
	EPC questionnaire	EPC is one-stop; EPCM manages the project for the owner	The same	Unchanged.
	17 Sep review	EPC does the work, EPCM manages the work; high-end MNCs engage EPCM		
IAQ's papers disagree	IAQ's documents		The site said	Now
	Discovery answers A2.3	1994	1995, on Nabilah's instruction of 10 Sep	1995 kept. IAQ's own papers still say 1994.
	Website copy sheet	Since 1994, 32 years		
	17 Sep review	Brand book corrected 1994 to 1995		
IAQ's papers disagree	IAQ's documents		The site said	Now
	Discovery answers A2.3	Malaysia, Singapore, Sweden, India, Germany	7 countries, on Nabilah's instruction of 10 Sep	7 kept, until IAQ confirms the list.
	Website copy sheet	8 countries		
Already agreed	IAQ's documents		The site said	Now
	Discovery answers A1.4	Semiconductor, Data Centre, EV Battery, Photovoltaics, District Cooling & Heating, Bio LifeScience, Food & Beverages	DV3's seven, in DV3's order	Unchanged. The copy sheet is the older list.
	Website copy sheet	Pharmaceuticals, six cards		

Fixed on the site	IAQ's documents		The site said	Now
	Feedback deck	This page is just too white	The first infographic set, white and red only	The set carries one bold colour per kind of thing, on a light ground, with soft glows.
	Moodboard	Keep the corporate colour, add one or two bold colours that complement it; hi-tech tone		
Already agreed	IAQ's documents		The site said	Now
	Discovery answers A2.1	We do not need to disclose subsidiary names, only the business model	No entity names on public pages	Unchanged. The deck title stays internal.
	Utility Solutions deck	Titled IAQ Utility Solutions		

Background

The competitor benchmark IAQ sent, the reading rules, every term placed by layer, and what is applied or next.

The benchmark: what Sunny expects.

On 18 September IAQ forwarded a competitor’s company introduction deck with one line: this is the expectation, infographics that help clients understand the business. Read slide by slide, it does six things our first Codex did not.

- 1

One idea per slide
The headline states it. Nothing else competes.
- 2

A drawing at the centre
Text sits round the drawing. No grid of text tiles.
- 3

Models inside the service picture
EPC (Engineering, Procurement and Construction) and EPCM (Engineering, Procurement and Construction Management) are named where they apply, not on a page of their own.
- 4

Proof coded the way a buyer classifies
By delivery type and discipline, with counts.
- 5

Every acronym spelled out
In a legend on the slide that uses it.
- 6

Numbers with an icon and a unit
Never a bare figure.

Their deck, slide by slide, and the IAQ slide that answers it

Their slide	The device	IAQ's answer		Where we stand
Group overview	Six facts with icons on the left; the three business units as columns on the right, each with its sectors.	Slide 1		Their columns carry share of sales. IAQ keeps revenue unpublished, so ours carry what each unit does, under IAQ's own photographs.
Global presence and performance	A world map, a workforce ring by function, three KPI tiles, two rings for sales by unit and by region.	Not in the set		Not in this set. Needs numbers IAQ has not released: headcount by function, the safety rate, order intake. The site's Global Presence page carries the map.
Industries and structure	An organisation tree: the group, its EPC arm with three business units and their sectors, a second arm for equipment and services, a heritage photograph.	Slide 1	Slide 2	Ours reads the three units along the life of the owner’s facility, which also says when each one is called.

Their slide	The device	IAQ's answer	Where we stand
Full spectrum of services	Five chevrons in two phases, the scope as bullets under each, the contract models named inside the stage they apply to, three claims at the foot.	Slide 5Slide 6	This is the slide that answers Nabilah’s 17 Sep point. Ours adds the business unit that carries each service, drawn under the bar.
Regional project record	A map with every project coded by delivery type, a count per type, and a legend that spells out each acronym.	Slide 11	Ours is a matrix, market by work delivered, computed from the registry. It becomes as strong as theirs when the approved project list arrives.
Positioning	One sentence, then three linked circles: end to end, predictable, proven.	Not in the set	The site hero and the closing band already say this. One slide if the profile wants it.
Value chain coverage	The client’s own process from front end to back end, with one proof project under each part.	Slide 2	Ours is the life of the facility rather than the life of the wafer, because IAQ sells to the facility owner.
Integrated installation	Capabilities and delivery value as two short lists beside a site photograph.	Slide 9	The unit pages on the site already do this with IAQ’s photographs.
Tool installation	A cutaway of a three-level fab, numbered callouts on the drawing tied to a numbered list beside it.	Slide 9	Ours is a section, with the base build in grey and what the hookup unit installs in red.
Modular solutions	Four numbered panels, a picture and three bullets each.	Not in the set	The unit pages’ step rails cover this; no slide needed yet.
Facility services lifecycle	A ring of services round a photograph, five numbers on the left, five pillars on the right.	Slide 10	Ours rings the six contract steps round the one number that matters to the buyer: no upfront investment.

The reading rules

EPC is a model, not a service. It governs how a project is delivered. The six services are the scope inside it. “EPC and Construction” repeats itself: EPC already includes construction.	A service can be bought alone. The model still governs it. Procurement only, design only, or design, procure and construct: each is a scope under the EPC model.
PCW and CDA are systems, not services. Process cooling water sits in mechanical work. Clean dry air sits in process work. Neither belongs in a service list.	Hookup is work. The contract can be EPC or EPCM. Tool hookup runs under either model, and it can be bought on its own.
EPC does the work. EPCM manages the work. The difference is who holds the construction contracts. Large multinational clients engage EPCM on the biggest programmes.	Engineers read labels, not descriptions. Procurement is “Procurement”. Commissioning is “Commissioning”. One line each, no paragraph repeating the word.

The terms, placed by layer

Models

EPC

Engineering, Procurement and Construction. The project delivery model, and the name of unit 1.

EPCC (Engineering, Procurement, Construction and Commissioning)

EPC plus Commissioning. IAQ delivers the whole project.

EPCM

EPC Management. IAQ manages the project; the owner holds the construction contracts.

ESCO

Energy Service Company, registered with Suruhanjaya Tenaga. The status EFM (Energy Facility Management) works under.

CaaS

Cooling as a Service. IAQ funds and runs the chiller plant for a fixed tariff.

Energy Performance Contracting

IAQ funds the upgrade and is paid from measured savings. Always spelled out on the site, never shortened to EPC.

Services

The six

Engineering design and consultation, Procurement, Construction, Testing and commissioning, Maintenance, Tools hookup.

T&C

Testing and Commissioning, service 4.

Tools hookup

Service 6, the work of connecting a tool. Never the name of a unit.

Units

EPC

Unit 1. Builds the facility.

Process Critical Utilities & Total Tool Installation Solutions

Unit 2, PCU and TTI for short. Re-equips a live semiconductor fab. Its deck is titled IAQ Utility Solutions; entity names stay off public pages.

EFM

Unit 3. Energy Facility Management. Runs and maintains the facility.

Work

CSA

Civil, Structural and Architectural.

MEP

Mechanical, Electrical and Plumbing. Also written M&E (mechanical and electrical).

Process

The utilities a production tool consumes: gases, chemicals, water, exhaust.

Systems

PCW

Process cooling water. Mechanical work.

CDA

Clean dry air. Process work.

UPW

Ultrapure water. Process work.

PV

Process vacuum. Process work.

HVAC, ACMV

Heating, ventilation and air conditioning; air conditioning and mechanical ventilation. Mechanical work.

FFU

Fan filter unit, the ceiling unit that makes a cleanroom clean.

BIM

Building Information Modelling, the coordinated 3D model every discipline is designed in.

Applied on the site

- The three business units lead the Services page; the six-service card set below them is gone, the cycle diagram carries the six with one line each. 17 Sep.
- The repeated “Start a project” band, the next-stage cards and the “where this goes next” strips are off every page. The closing band holds the one action. 18 Sep.
- Careers is one page: culture at the top, roles at the bottom, two entries in the menu. 18 Sep.
- The capability pack left the public Services page for the member portal. 18 Sep.
- The Services menu is three columns: the picture, the three units as cards (every section inside one a link, the contract models included), and the six services as one numbered list that lights what the unit under the pointer carries. No count or title block in any dropdown. 22 Sep.
- One name each, cross-checked against IAQ’s own documents: EPC; Process Critical Utilities & Total Tool Installation Solutions; EFM. 18 Sep.
- Unit 2 is semiconductor only and hook-up runs in IAQ’s own four phases, on the pages and the slides. 18 Sep.
- The member portal is its own area with tabs: Codex, Newsroom, Careers, Projects, Downloads. 18 Sep.
- Every unit page carries the scope and the work: the six services with the unit’s own lit, and the three disciplines with their systems. 18 Sep.
- The infographic set: a cover and eleven slides, one colour per kind of thing, on a light ground with soft glows, drawn to the benchmark IAQ sent on 18 Sep. Three of them are the unit extension pages for the company profile. 18 Sep.
- The closing band on every page is three sections: the ask with the contact lines beside it, the sitemap under its six main pages, and the sign-off. Careers uses the same band, pointed at HR. 22 Sep.

Next

- ❑ Place the three unit pages (slides 8, 9 and 10) in the company profile straight after the page on the three business units, and the rest in the order set out below. Shazwan.
- ❑ A responsive version of slides 2 and 5 for the public Services page.
- ❑ The cover and slides 2 and 9 on the booth screen beside the 3D model, for the 10 October backdrop and the 5 November screen content.
- ❑ The unit names in the company profile and the deck, to match the site.
- ❑ The 3D section on the home page shows the work by system type: civil and structural, architectural, process utilities, fire protection, air conditioning, electrical, cleanroom systems.
- ❑ The six service pages: keep them as depth reached from the cycle, or fold them into the Services page. IAQ to confirm.

3D archive.
The assembly the home page ran until 26 September 2026.

Brand Method's own scroll-driven build of IAQ's model: six stages, one accountable rail, the finish. Replaced on the home page by the developer's Scroll to build 3D (delivery "3D ONLY (4)", 26 September 2026), which assembles the same facility in 25 beats under IAQ's service headings and hands off to a first-person walkthrough. Kept here in full.



The facility, **layer by layer.**



Overall

Review of all eight systems · envelope closes · plant lands · site handover

9 / 9 · OVERALL

- Civil & Structural
- Architectural
- Process Utilities
- Fire Protection
- Air-conditioning
- HT & LV Electrical
- Cleanroom System
- Tools Hookup

Skip the build

